<u>Diabetes Technology Expert Program</u>

Glucose Sensors: the Master Module

1. Introduction

It's a pleasure to welcome you to this module focusing on glucose sensors, more formally known as continuous glucose monitors (CGMs).

In our modern age, individuals with diabetes are progressively substituting the traditional finger prick method with innovative glucose sensors. These devices have demonstrated a reduction in both hypoglycemia and hyperglycemia incidents, improved HbA1c levels, and better time-in-range. Thanks to these sensors, you gain a more comprehensive understanding of your glycemia as it records data every one to five minutes, which is significantly more frequent than the occasional finger prick method. This allows you not just to monitor your current glycemic levels, but also to predict the direction in which it's moving. The real-time alerts empower you to counteract hypoglycemia promptly, react quicker to hyperglycemia, and the feedback even encourages healthier eating and increased physical activity.

Therefore, glucose sensors are recommended for all insulin users, and can also be beneficial for individuals with diabetes and pre-diabetes who are not on insulin.

However, it's crucial to remember that there are certain caveats.

- There is a considerable range of sensor options, each with its own pros and cons. It's important to pick the most suitable sensor and utilize it correctly.
- It’s also vital to be aware of when to be skeptical of the glucose sensor readings and opt for a finger prick instead,
- understand which medications can interfere with sensor data,
- know what actions to take if allergic reactions or sensor detachment occur,
- and how to navigate through the extensive reports without getting overwhelmed.

These are precisely the topics this master module on glucose sensors will cover. The content has been written by Dr. Inge van Boxelaer, founder of Diabetotech and endocrinologist at the St-Lucas Hospital in Ghent in Belgium, and has been peer-reviewed by Dr. Liesbeth Van Huffel, endocrinologist at the OLV Hospital in Aalst in Belgium, and Melissa Holloway from the UK, a committed individual living with type 1 diabetes and founder of SmartStart Health.

Be assured, all the information provided here is completely independent from the sensor manufacturing companies, ensuring a genuine perspective on the advantages and drawbacks of different sensors.

Before we delve into the details, it is important to note that the information provided in these videos is not intended as medical advice. Always consult with your healthcare provider before making any adjustments to your diabetes therapy.

In the upcoming 40 minutes, you're going to delve into:

- The various types of glucose sensors available and their key differences
- Situations when a glucose sensor might not be reliable and a finger prick is necessary
- Proactive measures for scenarios involving allergies and/or sensor patch loosening
- And a straightforward method to interpret CGM reports effectively.
- Last but not least, we will highlight some great expert insights.

With this knowledge, you can confidently select the most suitable sensor for you or your patient, ensuring its proper usage for optimal diabetes management. Best of luck on this journey!

2. Comparison of different sensors

Welcome to this video detailing the assorted glucose sensors available today. We'll discuss the existing sensor types and their fundamental differences.

There are two types of glycemic sensors:

- flash glucose monitors, which display your glycemia when you flash or scan it with a reader or your mobile phone,
- and continuous glucose monitors that consistently transmit the sensor's glycemia to a receiver or mobile phone via Bluetooth.

2.1 Flash glucose monitor

Despite being the only type of flash glucose monitor, the FreeStyle Libre sensors are the most commonly used, largely due to its affordability. The widely used FreeStyle Libre 2 also includes adjustable high and low alarms.

Most current FreeStyle Libre 2 users now have access to an updated app that displays glucose values without the need to scan. Those using older FreeStyle Libre sensors or without the updated FreeStyle Libre 2 app must still scan their sensor manually.

These users are encouraged to scan the sensor every 8 hours at minimum to prevent data loss. At least 10 scans per day are recommended, as this correlates with improved glycemic control.

The sensor's glycemia is only visible upon scanning with a dedicated reader or the LibreLink app on your phone. Even if the alarm is triggered, a scan is required to view the glycemia.

The FreeStyle Libre sensors are unique in that they measure glucose levels every minute instead of every 5 minutes, which leads to more accurate alarms and glucose data.

Additionally, the FreeStyle Libre sensor offers the convenience of a 14-day wear period and combines the sensor and transmitter into one device, making it user-friendly.

If you want to access reports, you can utilize LibreView, a web-based program that allows you to view and analyze your glucose data.

2.2 Real-time / continuous glucose monitors

There are multiple continuous glucose monitor options.

- The Dexcom G6 providing a plethora of alarm settings and customization options. It is the most commonly used sensors in automated insulin delivery systems. The sensor can be worn for 10 days, and the transmitter has to be reused for 3 months.

- The Dexcom G7, successor to the Dexcom G6, functions as a combined sensor and transmitter. It offers a shorter warm-up time, a delayed high alarm, and a slightly longer duration of 10.5 days. A 15-day version of the Dexcom G7 will also be introduced, extending use to roughly 15.5 days.
- The Dexcom ONE Plus is similar to the Dexcom G7. However, the Dexcom ONE Plus sensor does not offer predictive alerts and cannot be linked to an automated insulin delivery system.
- The FreeStyle Libre 2 Plus is the same as the FreeStyle Libre 2, except that you no longer need to scan the sensor, and that the sensor life is 15 days instead of 14 days. This sensor is authorised to be used in combination with automated insulin delivery systems.
- The FreeStyle Libre 3 Plus is a full updated version of the FreeStyle Libre 2, and is also a continuous glucose monitor that no longer requires scanning of the sensor. It is the smallest and most accurate sensor available worldwide if you look at the MARD value, although all discussed sensors are accurate enough to dose insulin on.
- The Guardian sensors, developed by Medtronic, last for 7 days and the transmitter is reusable for a year. The Guardian 3 sensor typically requires two daily finger pricks for calibration, although three are recommended to minimize potential disruptions at inconvenient times. On the other hand, the Guardian 4 sensor only requires a finger prick for initiating the auto mode if you use the connected MiniMed insulin pump. It's worth noting that placing the Guardian sensor may require a bit more effort compared to some other sensors and requires considerably more overtape.
- The Simplera sensor is the successor of the Guardian sensors. It’s a sensor and transmitter in one device, that can be placed on the body with an easy inserter. No more overtape is needed. Just like the Guardian sensor, the sensor can be used for 7 days. There is a separate version to be used with and without the connected MiniMed insulin pumps.
- Medtronic has also started the rollout of the Instinct sensor, which is a version of the FreeStyle Libre 3 Plus that can connect to the MiniMed 780G insulin pump.
- Other lesser-known sensor manufacturers include Roche Diagnostics, Sinocare, Sibionics, MicroTech Medical, Medtrum, Yuwell, Syai Health, Ottai Medical, Bionime, i-Sens, Infinovo and Hippo Medical.

- There's also an implantable sensor available, the Eversense, which stays under the skin for 1 year. However, it does need an external transmitter and daily to weekly finger pricking for calibration.
- More than 100 invasive and non-invasive sensors are currently under development. A complete list with key specifications for each sensor can be found in the download section.

2.3 Do's & don'ts

When considering which sensor to use, several key aspects should be taken into account:

- Reimbursement and Cost: It is important to always check the availability of sensors in your country and understand the reimbursement options and financial implications. Without reimbursement, the cost of sensors can be quite high.
- Age Restrictions: Pay attention to the age restrictions associated with each sensor. Some are approved for use in adults only, while others can also be used in children of certain ages. However, if the benefits significantly outweigh the risks, these sensors are occasionally used off-label in younger children with diabetes.
- Automated Insulin Delivery Systems: If you're considering an automated insulin delivery system, make sure the sensor you choose is compatible with the system you're interested in.
- Allergies: If you're allergic or become allergic to the adhesive used in a sensor's patch, you may need to switch to a different brand.
- Alarm Settings: Most sensors come with adjustable high and low alarms. It's strongly recommended to activate the low alarm, typically set at 70 mg/dl or 3.9 mmol/l, or 60 mg/dl or 3.5 mmol/l for pregnant women. To avoid alarm fatigue, the high alarm is usually deactivated initially, but can be adjusted gradually based on your needs. Some sensors offer advanced alarm features such as predictive alarms, snooze functions, and customizable alarm tones. However, it's vital to only set alarms that will prompt you to take action; otherwise, you risk becoming desensitized to them over time.
- Data Sharing Options: Most sensors allow users to share their sensor data with family and caregivers in real-time. This feature holds significant value in various life situations. For parents and guardians, it offers reassurance and peace of mind, especially in classroom settings, as they can stay connected and informed while granting young adults their desired independence.

Moreover, data sharing acts as a crucial safety net for individuals living alone or working in demanding environments, providing an extra layer of security. Additionally, older adults benefit from the support of their loved ones through data sharing, ensuring their well-being and helping to mitigate potential health risks.

- Swimming and Showering: Sensors are typically water-resistant, which means you can wear them while showering or swimming. However, wearing a sensor in a hot tub, jacuzzi or while diving, is generally not recommended.
- Medical Imaging Procedures:
  - Remove the sensor before any MRI scan, except for the FreeStyle Libre 2, 3, and Plus versions, which are FDA-approved for use during MRI.
  - For RX and CT-scans, sensors can usually be left on. If possible, they can be shielded with a lead apron. After RX and CT scans, it is recommended to check the accuracy of the sensor with a finger prick.
- Air Travel: When flying, don't forget to bring a custom certificate as proof that you require this technology for your diabetes management. For airport security screenings, it's recommended to avoid wearing the sensors under full-body scanners and to request an alternative type of screening instead. Metal detectors at airports do not pose a problem for glucose sensors. Additionally, it’s important to carry the sensors in your carry-on luggage. If you check in your sensors, there is a risk of them freezing in the luggage compartment, which can damage them. If possible, avoid the baggage scanner for your carry-on luggage, because it uses X-rays.

The evolution in sensor types goes very fast. This content is updated in November 2025. For the latest updates, please visit the company's websites.

Additionally, staying updated with resources like the Diabetotech blog can help you explore the evolving diabetes tech field. Your perseverance, continual learning, and ability to adapt are integral to succeeding in the ever-evolving journey of diabetes management.

3. Reliability of glucose sensors

In this segment, we'll delve into the reliability of glucose sensors. We'll dissect the accuracy of sensors, trend arrows' interpretation based on lag time, instances where a finger prick is required, recommendations for sensor calibration, interfering substances and some extra tips and tricks.

3.1 Accuracy of glucose sensors

The reliability of glucose sensors like FreeStyle Libre, Dexcom, and Guardian sensors has significantly improved over time, with their Mean Absolute Relative Difference (MARD) consistently scoring below or around 10%. The MARD is widely debated as an accuracy parameter, yet sensors with an MARD of 10% or less are generally accepted to be sufficiently accurate for insulin dosing without requiring additional finger prick checks.

For a deeper look into the accuracy of newer sensors and why MARD alone is insufficient, see John Pemberton's *GlucoseNeverLies* blog.

Hospital use and the accuracy of these sensors during specific situations such as dialysis remain topics of ongoing discussion. In hospitals, sensors are typically permitted but are checked regularly with a finger prick. Finger pricks are also advised in case of in hospital hypoglycemia and after treating hypoglycemia with glucose.

It's important to keep in mind that sensors may be less accurate during the first 24 hours. So, be cautious when taking insulin or consuming sugars based on your sensor readings during this time.

3.2 Lag time and trend arrows

Glucose sensors measure glucose levels in the interstitial fluid within the fatty tissue below the skin. There is usually a delay, known as 'lag time', between changes in blood glucose and the sensor's detection of these changes. In general, the lag time ranges from 5 to 10 minutes but can extend to 15 to 20 minutes during times of rapid glucose change

This lag time plays a critical role in how we interpret trend arrows. Interpreting trend arrows on your glucose sensor is crucial in diabetes management. The specific details vary across systems, but typically, a straight up or down arrow indicates that the sensor glucose level is rising or falling approximately 2 mg/dl per minute or 0.11 mmol/l per minute. Numerous guidelines exist on how to adjust insulin based on these readings. However, one vital rule of thumb is that with a straight up or down arrow, the actual glucose level may already be about 30 mg/dl or 1.6 mmol/l higher or lower, considering the lag time.

Understanding this is vital to prevent overcorrection in instances of hypoglycemia and hyperglycemia.

- For instance, if you consume sugar due to hypoglycemia, there could be a delay of up to 20 minutes before the sensor reflects the rising glucose levels. In case you're unsure whether you need more sugar, it's advisable to perform a finger prick rather than solely depending on your sensor readings.
- Similarly, in cases of hyperglycemia after insulin injection, your glucose levels might already be dropping even if it's not yet noticeable on the sensor. If the trend arrow shows a downward trend, it's generally best to avoid further insulin injection. In case of uncertainty, performing a finger prick is recommended.

3.3 Situations warranting traditional finger pricks

Despite the overall reliability and convenience of glucose sensors, certain situations still necessitate the use of traditional finger pricks to confirm blood glucose levels.

- For instance, when your physical symptoms seem inconsistent with your sensor's readings, for example a hypoglycemia alert without symptoms of low blood glucose, it's crucial to conduct a finger prick for clarity.
- The same holds true when your sensor isn't providing a glucose value or a trend arrow.

The rule of thumb to remember here is, "when in doubt, get your meter out."

3.4 Finger pricks for validating sensor accuracy and calibration

When you notice a discrepancy between the sensor glucose value and your finger prick reading, it's important to consider a few factors before concluding that the sensor is malfunctioning. Here are some recommendations to keep in mind:

- Don’t do an accuracy check on the first day you’ve inserted a new sensor. As a rule, all sensors are likely to show a lower degree of accuracy on their first day, hence finger prick verification should be carried out more promptly.
- Rather than relying on a single reading, it's advisable to cross-check your sensor's accuracy multiple times before concluding its malfunctioning.

- Opt for a time when your glucose levels have been steady for 30 minutes (indicated by a flat arrow on your sensor), and also within range, that is between 70-180 mg/dl or 3.9-10 mmol/l.
- Wash your hands before you carry out a finger prick.

In case your sensor continually demonstrates more than 20% difference from finger prick readings under stable and target-range glucose conditions, consider calibrating it if it supports this feature, or replacing it with a new sensor. When contacted, most companies will typically provide a replacement sensor free of charge.

3.5 Interfering substances

It's important to understand that sensors can be affected by certain interfering substances. This varies depending on the sensor you use.

- While these are typically infrequent substances, even common ones like paracetamol, also known as acetaminophen, can interfere with sensor readings. Dexcom sensors only show this interference when the daily dosage exceeds 4 grams, which is the maximum safe dosage for adults. However, lower doses of paracetamol can result in falsely high readings with the Guardian sensor. Thus, if you're taking paracetamol while using a Guardian sensor, an additional finger prick is recommended for cross-verification.
- Most FreeStyle Libre sensors may show falsely elevated readings when high doses of vitamin C, specifically more than 500 mg per day ascorbic acid, are consumed. It's important to note that while this scenario is not frequently observed in Europe, if a patient mentions high vitamin C intake, it becomes an important factor to consider.
- Both Dexcom and Guardian sensors have reported false highs when used in conjunction with Hydrea, a medication used to treat malignant blood diseases. This is a rare situation but an important one to keep in mind if Hydrea appears on a patient's medication list.

3.6 Do's & don'ts

- Many individuals, when they first start using a sensor, are alarmed by the spikes in sensor glucose following a meal. They might feel inclined to administer a corrective bolus of insulin immediately. It's crucial to remember, however, that there is still active insulin working in the body after the meal, and administering extra insulin too hastily can result in insulin stacking - an undesirable scenario that can lead to severe hypoglycemia.

- A common surprise for many people is the significant difference between sensor readings and those obtained from a finger prick. It's essential to understand that a sensor's accuracy can best be evaluated at times when sensor glucose levels have been stable and within the target range. Several checks are required before concluding that a sensor isn't functioning properly.

- Most sensors are calibrated at the factory, but they need a warm-up period of 30 minutes to 2 hours before they start displaying glucose values. If there are significant fluctuations in sensor glucose during this warm-up period, the sensor's data may contain errors. Consequently, it's wise to replace your sensor at a time when you anticipate stable blood glucose levels - not immediately before or after a meal.

While glucose sensors can be relied upon for insulin dosing, it may take some time for individuals to fully trust the sensor readings. It is important to acknowledge that all technology can have failures, and glucose sensors are generally less accurate than finger pricks. If your sensor reading does not match how you feel, always verify your glucose level with a control finger prick. By following these guidelines, glucose sensors can be a reliable tool for insulin dosing and can significantly ease the management of diabetes.

In the next chapter of this course, we will cover how to ensure the proper adhesion of the sensor and manage potential allergic reactions.

4. Effective sensor site management

Welcome to this video on effective site management.

In this segment, we'll delve into some useful tips regarding sensor placement, removing sensors, sensor allergy, and provide a list of do's and don'ts for sensor maintenance.

4.1 Optimal sensor site management

The accuracy of sensor readings can significantly depend on where they're placed:

- The optimal results with FreeStyle Libre sensors are achieved when they are positioned on the back of the upper arms. Some individuals use them in alternative off-label locations. However, it has been observed that placing the FreeStyle Libre sensor on the abdomen resulted in lower accuracy compared to the thigh, which, in turn, was slightly less accurate than the upper arm placement. Dexcom sensors perform optimally on the upper arms but can also be placed on the abdomen. For those under 18, buttocks placement is also permissible.
- Guardian sensors similarly provide top accuracy on the upper arms and can be positioned on the abdomen. Practically, many users also place them on the thigh, as this seems to cause less bleeding in some people.
- Avoid repeated use of the same sensor site to prevent local skin inflammation. Change the sensor site with each new sensor.
- Steer clear of placing the sensor under a belt, waistband, safety belt, near bones, the navel, irritated skin, scars, tattoos, or areas of lipodystrophy.
- Be wary of areas prone to bumping, pushing, or laying on while sleeping. Pressure on the sensor during sleep can cause compression, leading to false low readings.

4.2 Ensuring sensor longevity

To make your sensor adhere better, try the following tips:

- Clean and dry the skin thoroughly. Shaving and washing with soap and water can remove oils that prevent proper adhesion. Allow the area to air dry before sensor application.
- Avoid applying sunscreen or insect repellent around the sensor location.
- To improve sensor adhesion, it may be beneficial to disinfect the skin prior to sensor placement. Wait until the skin is dry before inserting the sensor. If the skin is still moist from the alcohol wipe, the sensor may not adhere properly. It is worth mentioning that using alcogel for disinfection is not recommended as it can affect the accuracy of the sensor readings and cause values to drift.
- In the summer, consider applying deodorant first to minimize localized sweating.

- Some people find using additional glues like Skintac or Conveenprep beneficial. Let the adhesive dry completely before sensor insertion. On the other hand, some people may have an allergic reaction to Skintac, and this is sometimes mistakenly associated with an allergy to the sensor itself rather than the wipes.
- After sensor placement, make sure to rub the adhesive well onto the skin to enhance its adherence.
- Dexcom and Guardian sensors typically come with free overtape included in the package. FreeStyle Libre and Simplera sensors usually do not require additional overtape.
- Several companies offer overtape with custom sizes, adorable designs, and high-quality materials. However, it is sometimes recommended not to cover the transmitter to avoid potential interference with Bluetooth connectivity.
- If you place a small piece of tissue between the sensor and the overtape, you can easily replace the overtape without having to replace the sensor itself. This method can provide convenience and flexibility in maintaining the sensor's adhesion.
- If the side of the overtape starts to come off, you can use long-form medical tape to secure it in place.
- Additionally, there are sensor holders available that can effectively keep the sensor securely positioned during daily activities.

4.3 Careful sensor removal

Some users find their sensors adhere too securely. Here are some tips for sensor removal:

- Remove the sensor slowly, keeping it parallel to the skin as much as possible.
- You may use medical adhesive solvent or cotton wool soaked in baby oil around the sensor. This can help make it easier to remove the sensor.
- Avoid sensor removal post-shower. Heat and humidity enhance adhesive stickiness, making it harder to remove.
- Adhesive residues can be removed using specialized products like Remove wipes, Tacaway, or Limisan plaster remover. Baby oil or massage oil can also be used.

4.4 Handling sensor allergies

Up to a third of sensor users may experience some form of skin reaction. It is crucial to ask about these reactions and provide appropriate tips to ensure optimal sensor use. Here are some recommendations:

- Cleanse the skin thoroughly: Prior to sensor insertion, ensure the skin is properly cleaned. If you are prone to irritation, it is advisable to avoid alcohol wipes as they can exacerbate the reaction. Instead, consider using barrier wipes or spray, allowing this to dry adequately before placing the sensor.
- Consider corticoid spray pre-insertion: If skin reactions persist, you can try using cortisone nasal spray before sensor insertion. Allow these substances to dry thoroughly before placing the sensor on the skin.
- Explore alternative patches: Allergies often stem from the adhesive patches used with the sensor. If you experience ongoing irritation, switching to a different type of patch may be beneficial. Experimenting with alternative options can help identify a patch that suits your skin better.
- Use cortisone ointment for irritation: In cases where irritation occurs, applying cortisone ointment can help expedite skin healing.
- Finally, consider using a lipid-rich cream on the sensor sites after sensor removal, to make sure that the skin can heal faster. A dry skin is more prone to irritation.

4.5 Contact irritation versus contact allergy

When addressing allergies, it's important to assess the severity of the reaction and examine any visible rash. While most cases involve contact irritation, a small percentage of sensor users may experience contact allergy. In such instances, it's crucial to avoid the allergen to prevent further complications.

- Contact allergy is an immune system hypersensitivity response triggered by an adhesive component. It manifests as a delayed allergic reaction occurring weeks, months, or years after initial contact. Subsequent exposures lead to an immediate reaction within 1 to 2 hours, intensifying with each occurrence. Symptoms include intense itching, vesicles, papules, skin swelling, and oozing fluid. The reaction may extend beyond the exposed area.
- Contact irritation, on the other hand, results from non-immune mediated inflammation due to direct skin damage caused by chemical or mechanical factors. Heat, humidity, and exposure duration can worsen the symptoms, which typically include a burning sensation, redness, peeling, and crusting.

- Distinguishing between contact allergy and irritation can be challenging, and they can occur together. Skin damage from irritation can increase antigen exposure, potentially triggering an allergic reaction. If needed, it is advisable to consult a specialized dermatologist for further evaluation and guidance in such cases. They can provide expert insight and help determine the appropriate course of action.
- Conducting a patch test can aid in distinguishing between the two conditions. This involves evaluating common allergens like colophonium, isobornyl acrylate (IBOA), and methyl acrylates. While some device manufacturers disclose known allergens, they sometimes change adhesive formulations without notification.

4.6 Do's & don'ts

- If you experience persistent discomfort after sensor insertion, it could be in contact with a nerve or muscle. It's best to remove it and insert a new one.
- When inserting a sensor, it is recommended not to press the inserter too firmly in order to avoid bleeding. The sensor wire should not be inserted into a blood vessel, so if the sensor site bleeds profusely it's appropriate to change the sensor and replace it, and report to the company to obtain a free-of-charge replacement sensor. If it is only a minor bleeding, you can try to keep the sensor.
- If recurring sensor errors occur try selecting a different body site to set your sensor.
- Bluetooth connection may be disturbed by other nearby Bluetooth devices such as blood glucose meters, headsets, tablets or kitchen devices such as microwave ovens or ceramic hobs. In this case your receiver will not display any sensor glucose values. When the bluetooth connection is re-established, the data is backfilled.

Armed with this valuable information, you are now equipped to handle common sensor issues, including optimal sensor placement, secure removal, addressing loosening, and managing allergies. With this understanding, you can harness the power of glucose sensors to effectively manage diabetes and take control of your health.

<u>5. Interpreting sensor reports</u>

Welcome! In this video, we'll delve into the fascinating world of sensor reports, equipping you with the knowledge and skills to interpret and navigate through the valuable data they provide. Congratulations on making it this far in the module!

Now, let's dive into interpreting sensor reports. These reports hold the key to unlocking valuable insights into glucose control and optimizing diabetes management. Whether you're a healthcare provider or an individual with diabetes, this information will empower you to make informed decisions and take charge of your journey towards optimal health.

5.1 Types of reports

As you explore sensor reports, you'll encounter a variety of formats. From summary reports to daily reports and device settings, each sensor manufacturer brings its own unique approach to presenting data. While this may initially seem overwhelming, fear not! We'll guide you through the intricacies and provide the tools you need to navigate these reports

When reviewing reports, you have the flexibility to choose the desired period, typically looking at the last 2 weeks.

Most sensor manufacturers, excluding Medtronic, also provide an Ambulatory Glucose profile (or AGP) report—a global effort to standardize sensor reports.

Within the AGP report, you'll find valuable information presented in a user-friendly manner:

- At the top, international guidelines on time in range, time below range, and glucose variation are displayed, along with the person's actual glycemic control. A color bar on the left further enhances clarity.
- Below that, you'll see trends from the chosen period.
- Further down, trends by weekday are presented, providing deeper insights into glycemic patterns.

5.2 Interpreting reports

Now, let's delve into the art of interpreting reports from a stand-alone sensor:

- Step 1: Begin by checking if the sensor has been worn for more than 90% of the time. If not, discuss any challenges that may hinder consistent wear.
    - If there are issues with supplies or sensors not lasting the full sensor wear time, contact the manufacturer for replacement sensors.
    - For individuals experiencing skin problems or difficulty keeping the sensor in place, explore strategies like rotating insertion sites and utilizing barrier products, overtapes, or adhesive removers to protect the skin.
- Step 2: Assess if the user is meeting their glycemic targets by examining the top panel of the AGP profile, which displays ten core parameters.
    - Remember, positive feedback plays a crucial role in supporting self-empowerment. Recognize and celebrate progress, such as improvements in time in range, to encourage individuals in their diabetes management journey.
    - Then look for any areas of concern or opportunities for improvement.
        - Time in range should ideally be above 70% for most people with diabetes. Even a 5% improvement in TIR represents a clinically relevant outcome, as this is equivalent to a 0.3% decrease in HbA1c.
        - Time below range should be less than 4%. It is better to talk about minutes instead of percentages as this is easier to understand (1% equals 15 minutes).
        - Glycemic variation is best kept below 36%, although this may be easier for individuals with type 2 diabetes compared to those with type 1 diabetes. A glycemic variation above36% signifies an unstable glycemia and a higher risk of hypoglycemia.
- Step 3: If any issues arise, turn to the AGP profile for a visual representation of where the problem lies.
    - Prioritize the treatment of hypoglycemia, aiming for "more green and less red" in the graph.
    - Strive for a flat, narrow, and in-range curve, and analyze patterns of hypo- and hyperglycemia.
    - Before making any adjustments, review the bottom panel of the AGP report, which displays daily patterns. This analysis helps rule out whether patterns are exclusive to specific work or rest days.

    - Additionally, examine the daily reports to ensure correct timing of insulin boluses and check for systematic overcorrection of hypo- or hyperglycemia.
- Step 4: Ensure that the used alerts align with the specific needs of the individual, to prevent alarm fatigue. Customizing alarms helps maintain a healthy balance between staying informed and avoiding unnecessary disturbances.
- Step 5: Set a single goal collaboratively and plan the next consultation for review. Focus on improving areas of concern—whether they relate to lows, highs, or wide glycemic fluctuations.

Interpreting sensor reports can feel overwhelming due to the abundance of glucose data available. To manage this effectively, focus on addressing the most noticeable trends rather than attempting to fix everything at once. This approach helps in gradually improving glucose control step by step. Aim to provide transparent and clear instructions based on the sensor reports, empowering patients to interpret and act on the data independently. A helpful tip is to print out the AGP profile and summarize the therapy changes on paper, allowing patients to review their results and conclusions even after the consultation.

Congratulations on reaching this stage of the module! You now know how to interpret sensor reports, which will help you to provide excellent diabetes care and make a significant impact on the lives of those you care for.

6. Expert insights

Welcome to the final chapter of our comprehensive module on glucose sensors! You've come a long way, acquiring valuable knowledge and skills to enhance your expertise in diabetes care. Congratulations on reaching this milestone!

In this concluding segment, we invite you to watch an engaging YouTube video that humanizes glucose sensors and offers insights from various experts. This video explores different CGM options such as Freestyle Libre, Dexcom, Eversense, and Medtronic. The experts discuss the pros and cons of each CGM and share their personal preferences. It's important to note that there is no one-size-fits-all CGM, and it's crucial to consider the advantages and disadvantages before making a decision.

https://www.youtube.com/watch?v=fng5McLfVUo (14.5 min)

YouTube is a fantastic platform for discovering fun and interesting tips and tricks about glucose sensors. Although these channels are not regulated, they are managed by individuals living with diabetes themselves. Their content truly adds value to the diabetes community.

Now, you've reached the end of the module on glucose sensors. You have expanded your knowledge about various sensors, learned about potential pitfalls such as when to perform a finger prick, and substances that can interfere with sensor accuracy. You've gained valuable tips on sensor adherence and managing allergies. Additionally, we explored how to interpret glucose sensor reports and highlighted a cool YouTube channel for further education and inspiration. Your expertise has undoubtedly grown, enabling you to troubleshoot effectively when using glucose sensors. We hope you found this course enlightening and inspiring.

Remember, your learning journey doesn't end here. Continue to explore the other modules on specific glucose sensors, insulin pumps and automated insulin delivery systems, and embrace the spirit of lifelong learning.
Once again, congratulations on your remarkable achievement in completing this module on glucose sensors. We wish you continued success and fulfillment in your diabetes care endeavors. You are making a real difference!